Daily Streamflow Forecasting

MLP and LSTM | Leaf River near Collins, Mississippi

Finding. On the untouched 2011-2012 test period, the feed-forward MLP had the lowest overall error (2.748% of maximum flow). The LSTM was close (2.824%). Both improved on persistence and linear regression.

1. Problem and objective

This project predicts the Leaf River's daily discharge from recent rainfall and river flow. It compares a single-hidden-layer feed-forward neural network (MLP) with a sequence-based Long Short-Term Memory network (LSTM). Persistence (tomorrow equals yesterday) and linear regression are simple reference models. The experiment adapts Xiao and You (2020), adding a separate test period and explicit baselines.

2. Dataset and independent cross-check

Daily discharge is from USGS station **02472000** (Leaf River near Collins, Mississippi); precipitation is from NOAA GHCN-Daily station **USC00221852** (Collins, Mississippi). The period is 1 January 2003-31 December 2012: **3,653 daily rows**. There are 46 missing rainfall values (1.3%) and no missing flow values. Maximum flow is 29,700 cfs; maximum daily rain is 6.96 in.

The supplied daily CSV was compared row-by-row with a fresh merge of the supplied raw USGS and NOAA files, selecting station USC00221852. Dates and both measurements matched exactly. The 6.96-in maximum matches the project guide's expected rebuild; it is slightly above the 6.8-in maximum reported in the reference paper, so the rainfall records are close but not asserted to be identical.

3. Experimental design

The split is chronological to keep future observations out of training: **2003-2007 train**, **2008-2010 validation**, and **2011-2012 test**. Each prediction uses lagged rainfall and flow. Inputs are scaled using training-period maxima (flow: 29,700 cfs; rain: 6.96 in). Models use Adam and mean squared error, with early stopping on validation loss. The quick search evaluated 48 MLP and 12 LSTM settings; selected neural configurations were retrained with five random seeds.

Period	Years	Role
Train	2003-2007	Fit model weights
Validation	2008-2010	Select settings and stop training
Test	2011-2012	Final comparison only

Model selection and inputs

For each date t, the target is discharge Q(t). The MLP receives a flat vector of preceding rain and flow values; the LSTM reads the preceding daily rain-flow pairs in time order. The grid selected settings using validation RMSE, and the final test period remained untouched until evaluation.

Model	History used	Selected size
MLP	5 rainfall + 5 flow days	8 hidden units
LSTM	2 daily rain-flow pairs	12 LSTM units
Baselines	Persistence; linear regression on MLP inputs	No neural hidden layer

4. Results: compare on the test set

Use test scores for the final comparison; validation scores selected settings. RMSE, peak RMSE, non-peak RMSE, and peak bias are percentages; lower error is better. NSE is unitless; higher is better. Peak days have observed flow above 1,500 cfs. Neural-network RMSE is mean plus/minus standard deviation across five seeds.

Model	RMSE %	Peak RMSE %	Non-peak %	NSE	Peak bias %
MLP	**2.748 +/- 0.094**	8.165	0.696	0.815	-3.25
LSTM	2.824 +/- 0.034	8.400	0.589	0.797	-6.63
Persistence	3.232	9.649	0.755	0.744	-0.13
Linear regression	3.055	9.116	0.721	0.771	-9.11

Selected settings: MLP 5 rain lags, 5 flow lags, 8 hidden units; LSTM 2-day sequence, 12 hidden units. Run metrics agree with supplied reference results to rounding.

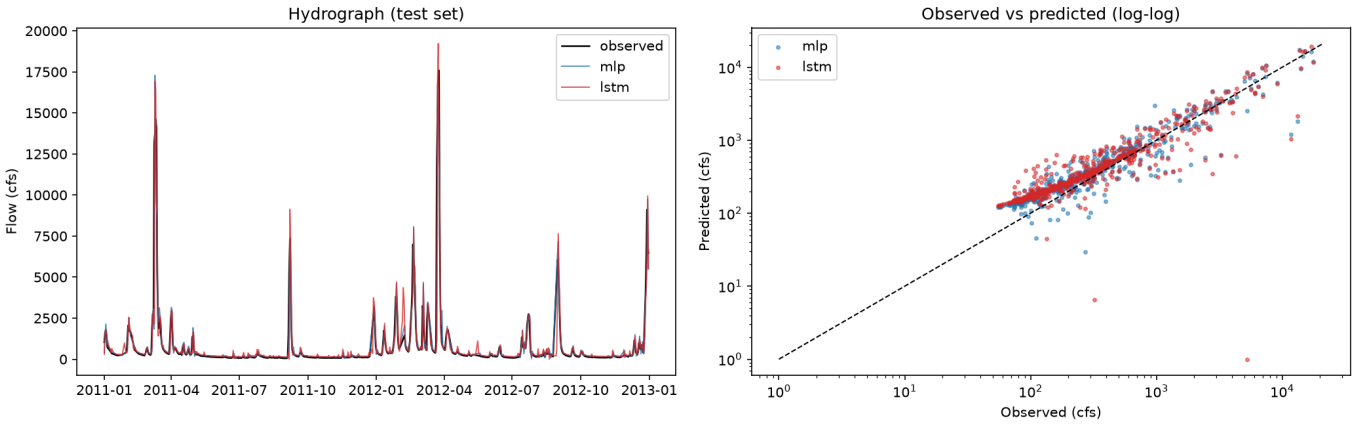


Figure 1. Test-period hydrograph (left) and observed-versus-predicted discharge (right). The dashed diagonal marks perfect predictions; right-hand axes are logarithmic.

5. Interpretation and conclusion

The MLP reduced test RMSE by 15.0% versus persistence and 10.0% versus linear regression. It had the best overall test RMSE and NSE (0.815). The LSTM had lower non-peak error (0.589% versus 0.696%), but slightly higher overall error and stronger average underprediction on peak days (-6.63% versus -3.25%). Peak errors were much larger than non-peak errors for both networks, so sudden floods remain the main challenge.

The results support one-day-ahead prediction from recent rainfall and flow on this dataset, with the MLP as the strongest overall model in this run. They do not establish operational flood-warning performance: the study covers one river and one rain gauge, uses a short look-back, and evaluates a two-year test period.

References: Xiao, M. & You, S. (2020), *Application of Artificial Neural Network in Streamflow Forecasting*, Stanford CS229 project. Data: USGS station 02472000; NOAA GHCN-Daily station USC00221852. Assignment deliverable requirements: supplied UE24CS352A mini-project guidelines.